Experiment 8

Round Robin Schedulingcode:

```
#include <stdio.h>

int main(void) {
   int n, q, i, time = 0, remain;
   int bt[20], rem[20], wt[20], tat[20];
   float sw = 0, st = 0;

   printf("Number of processes (max 20): ");
   scanf("%d", &n);

   if (n < 1 || n > 20)
      return 1;

   for (i = 0; i < n; i++) {
      printf("Burst time for P%d: ", i + 1);
      scanf("%d", &bt[i]);
      rem[i] = bt[i];
   }

   printf("Time quantum: ");
   scanf("%d", &q);

   if (q <= 0) {
      printf("Quantum must be positive.\n");
      return 1;
   }

   remain = n;
```

```
while (remain > 0) {
   int progressed = 0;

   for (i = 0; i < n; i++) {
      if (rem[i] > 0) {
         int slice = rem[i] < q ? rem[i] : q;

         rem[i] -= slice;
         time += slice;
         progressed = 1;

         if (rem[i] == 0) {
            tat[i] = time;
            wt[i] = tat[i] - bt[i];
            remain--;
         }
      }
   }

   if (!progressed)
      break;
}

printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++) {
   printf("P%d\t%d\t%d\t%d\n",
         i + 1, bt[i], wt[i], tat[i]);

   sw += wt[i];
   st += tat[i];
```

```
    }

    printf("Average waiting time = %.2f\n", sw / n);
    printf("Average turnaround time = %.2f\n", st / n);

    return 0;
}

}
```

Output:

